

STAT 110 · Reasoning with Data

Meridian State University · Fall 2026 · 3 credits · 15 weeks (Aug 24 – Dec 11)

Instructor: Dr. Amara Okafor · a.okafor@meridianstate.example · Office hours Tue 2–4 pm, Hale 214

Meets Tuesday and Thursday, 9:30–10:45 (75 min), Hale 110. Prerequisite: none. 100-level.

Course description

An introduction to reasoning with data for students who will read statistics far more often than they compute them. We ask statistical questions, collect and describe data, compare groups, and judge claims made in public.

Required text

OpenIntro Statistics, 4th edition (free online).

Learning outcomes

By the end of the course you will be able to:

1. Tell a statistical question from a non-statistical one.
2. Understand variability in data.
3. Describe the center and spread of a distribution.
4. Design a simple study to answer a question.
5. Communicate findings to a non-technical audience.
6. Critique a statistical claim in the news.

Grading

Component	Weight	Notes
Weekly quizzes	15%	Lowest two dropped
Homework sets	20%	Six sets
Midterm exam	20%	Week 8, in class, multiple choice
Course project	40%	Due Thursday, December 4
Participation	5%	

Final grades: A 90–100, B 80–89, C 70–79, D 60–69.

Course schedule

Week	Dates	Topic	Reading	Due
1	Aug 25–29	What is a statistical question?	Ch. 1	—
2	Sep 1–5	Cases, variables, data tables	Ch. 1	HW 1
3	Sep 8–12	Summarizing one variable	Ch. 2	Quiz 1
4	Sep 15–19	Center and spread	Ch. 2	HW 2
5	Sep 22–26	Shapes of distributions	Ch. 2	Quiz 2
6	Sep 29–Oct 3	Sampling and study design	Ch. 3	HW 3
7	Oct 6–10	Bias and confounding	Ch. 3	Quiz 3
8	Oct 13–17			
9	Oct 20–24	Comparing two groups	Ch. 4	HW 4
10	Oct 27–31	Sampling variability	Ch. 4	Quiz 4
11	Nov 3–7	Confidence intervals	Ch. 5	HW 5, project draft
12	Nov 10–14	Hypothesis tests	Ch. 5	Quiz 5
13	Nov 17–21	Reading a study	Ch. 6	HW 6
14	Dec 1–5	Communicating results	Ch. 6	Project due Dec 4

Course policies

Attendance: attend both weekly meetings; more than three unexcused absences lowers the participation grade.

Late work: 10% per day, up to five days. No late quizzes.

Academic integrity: the Meridian State honor code applies to all work.

Generative AI: you may use AI tools to check your reasoning; you may not submit AI-written text as your own. Cite any AI use.

Accommodations: contact Accessibility Services in the first two weeks to arrange accommodations.